

IMP TOPICS OF BIOLOGY BY PRAGYA

Based on the analysis of the six PDF sources provided, which are Class 12 Biology question papers (Sets 57/5/1, 57/7/1, 57/1/1, etc.), here are the most important topics for every chapter. These topics appear frequently across the different sets as multiple-choice, short-answer, long-answer, or case-based questions.

Unit VI: Reproduction

Chapter 2: Sexual Reproduction in Flowering Plants

- **Pollination:** Mechanisms and agents, particularly **Co-evolution** (e.g., Fig & Wasp, Orchid Ophrys & Bee) and water pollination (e.g., Vallisneria, Zostera).
- **Embryo & Endosperm Development:** The stages of development in dicots and monocots, and the concept that endosperm development precedes embryo development.
- **Gametophyte Structure:** Development of the **Male Gametophyte** (pollen grain) and the **Female Gametophyte** (Embryo Sac structure including synergids, antipodals, and egg apparatus).
- **Artificial Hybridization:** Steps like Emasculation and Bagging.
- **Outbreeding Devices:** Mechanisms to encourage cross-pollination.

Chapter 3: Human Reproduction

- **Menstrual Cycle:** Detailed hormonal control (LH, FSH, Progesterone, Estrogen) and the events in the ovary and uterus (proliferative, luteal phases).
- **Gametogenesis:** The process of **Spermatogenesis** (hormonal control, structure of sperm) and **Oogenesis** (formation of follicles, polar bodies).

- **Fertilization & Implantation:** The interaction of sperm and ovum (acrosomal reaction, block to polyspermy) and the formation/function of the **Placenta**.
- **Parturition:** The neuroendocrine mechanism involving signals from the fully developed foetus and Oxytocin.

Chapter 4: Reproductive Health

- **Assisted Reproductive Technologies (ART):** Procedures like **IVF**, **ZIFT**, **GIFT**, **ICSI**, and **IUI** used to treat infertility.
- **Contraception:** Mode of action of **Oral Pills** (hormonal) and **IUDs** (Intra Uterine Devices like Copper-T).
- **Amniocentesis:** Its use for detecting genetic disorders and its misuse for sex determination.
- **Sexually Transmitted Infections (STIs):** Examples and complications if left untreated.

Unit VII: Genetics and Evolution

Chapter 5: Principles of Inheritance and Variation

- **Chromosomal Disorders:** Karyotypes and symptoms of **Down's syndrome** (Trisomy 21), **Turner's syndrome** (45, XO), and **Klinefelter's syndrome** (47, XXY).
- **Mendelian Disorders:** Pedigree analysis of Sex-linked recessive traits like **Haemophilia** and **Colour Blindness**, and Autosomal traits like **Thalassemia**.
- **Sex Determination:** Mechanisms in Humans (XY), Birds (ZW), and Insects/Grasshoppers (XO).
- **Inheritance Patterns:** **Co-dominance** (Blood groups ABO) and Multiple Alleles.

Chapter 6: Molecular Basis of Inheritance

- **DNA Replication:** The structure of the **Replication Fork**, continuous vs. discontinuous synthesis, and the role of enzymes (DNA ligase, Polymerase).
- **Transcription & Translation:** The process of making mRNA (Promoter, terminator) and protein synthesis (Genetic Code, start/stop codons).

- **Lac Operon:** Regulation of gene expression.
- **DNA Fingerprinting:** The principle (VNTRs/Mini-satellites) and applications.
- **DNA Structure:** Nucleosome model and Histone octamer.

Chapter 7: Evolution

- **Evidence of Evolution: Homologous organs** (Divergent evolution) vs. **Analogous organs** (Convergent evolution).
- **Natural Selection:** Industrial Melanism (moths in England) and types of selection curves (Stabilizing, Directional, Disruptive).
- **Adaptive Radiation:** Examples like Darwin's Finches and Australian Marsupials.
- **Human Evolution:** Stages from Dryopithecus to Homo sapiens, focusing on brain capacity and habits.
- **Hardy-Weinberg Principle:** Factors affecting equilibrium.

Unit VIII: Biology in Human Welfare

Chapter 8: Human Health and Disease

- **Common Diseases:** Life cycle of the malarial parasite (**Plasmodium**), causative agents and symptoms of **Typhoid**, **Pneumonia**, **Ascariasis**, **Ringworm**, and **Filariasis**.
- **Immunity: Antibody structure** (IgA, IgG), Innate vs. Acquired immunity, and Active vs. Passive immunity.
- **AIDS & Cancer:** HIV replication mechanism, properties of cancer cells (loss of contact inhibition), and detection techniques (Biopsy, MRI).
- **Drugs & Alcohol:** Effects of **Opioids**, **Cannabinoids**, **Coca alkaloids**, and Tobacco.

Chapter 10: Microbes in Human Welfare

- **Sewage Treatment:** The function of **STPs** (Primary vs. Secondary treatment), formation of **flocs**, and activated sludge.
- **Bioactive Molecules:** Microbes producing **Cyclosporin A** (Trichoderma), **Statins** (Monascus), and Streptokinase.

- **Biofertilizers & Biogas:** Role of methanogens and organic farming.

Unit IX: Biotechnology

Chapter 11: Biotechnology - Principles and Processes

- **Restriction Enzymes:** Naming conventions (e.g., EcoRI), recognition sites (palindromes), and the formation of sticky ends.
- **Cloning Vectors:** Features of **pBR322** (ori, rop, antibiotic resistance genes like AmpR, TetR) and Insertional Inactivation (Blue-white screening).
- **Techniques: PCR** (Polymerase Chain Reaction) steps and **Gel Electrophoresis**.
- **Bioreactors:** Stirred-tank bioreactors for large-scale production.

Chapter 12: Biotechnology and its Applications

- **Genetically Modified Crops: Bt Cotton** (mechanism of Cry genes/toxins) and **RNA interference (RNAi)** for nematode resistance in tobacco.
- **Medicine:** Genetically engineered **Insulin** (Pro-insulin to Insulin) and **Gene Therapy** (Adenosine Deaminase - ADA deficiency).
- **Transgenic Animals:** Reasons for creation (e.g., Vaccine safety, Biological products like Rosie the cow's milk).
- **Biopiracy & Ethics:** GEAC and patent issues (Basmati rice, Turmeric).

Unit X: Ecology

Chapter 13: Organisms and Populations

- **Population Interactions: Mutualism** (Fig-Wasp), **Parasitism**, **Competition** (Gause's exclusion principle), and **Commensalism**.
- **Population Growth: Logistic** vs. **Exponential** growth curves (Verhulst-Pearl equation) and age pyramids.
- **Population Attributes:** Birth rate, death rate, and sex ratio.

Chapter 14: Ecosystem

- **Ecological Pyramids:** Pyramids of **Biomass**, **Number**, and **Energy** (upright vs. inverted examples like sea biomass).
- **Productivity:** Gross Primary Productivity (GPP) vs. Net Primary Productivity (NPP).
- **Decomposition:** Factors affecting the rate of decomposition.

Chapter 15: Biodiversity and Conservation

- **Biodiversity Loss:** The "Evil Quartet" (especially **Alien species invasion** and Habitat loss/fragmentation).
- **Conservation Strategies:** **In-situ** (Biodiversity Hotspots, Sacred Groves) vs. **Ex-situ** conservation (Cryopreservation, Seed banks).
- **Species-Area Relationship:** Alexander von Humboldt's observations and graph.

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